EFY TESTED Abhay Verma

A GPS-Enabled SMART AIR COMPRESSOR

Guessing tyre pressure is a gamble. Within just '6 hours', you can complete this medium-complexity project under ₹3200: a battery-powered ESP32 compressor that inflates to your target PSI, switches itself off, streams live pressure to a dashboard, and even tracks its GPS location.

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Tired of guessing tyre pressure? Over-inflation can damage tyres and balls, while under-inflation reduces performance and compromises safety. This project is a smart portable air compressor that measures pressure in real time, runs until your target pressure is reached, and then shuts itself off automatically.

The system is built around an ESP32 microcontroller, which handles power management, motor control, pressure monitoring, GPS communication, and the user interface. A single rechargeable 12V battery feeds two paths: one supplies the 12V DC compressor motor through a metal-oxide-semiconductor field-effect transistor (MOSFET) switching module, while the other passes through a DC-DC buck converter to give a regulated 5V for the ESP32, pressure sensor, and GPS module.

In operation, the pressure sensor sends the ESP32 an analogue voltage proportional to the measured pressure. When the reading falls below the preset lower limit, the ESP32 switches the

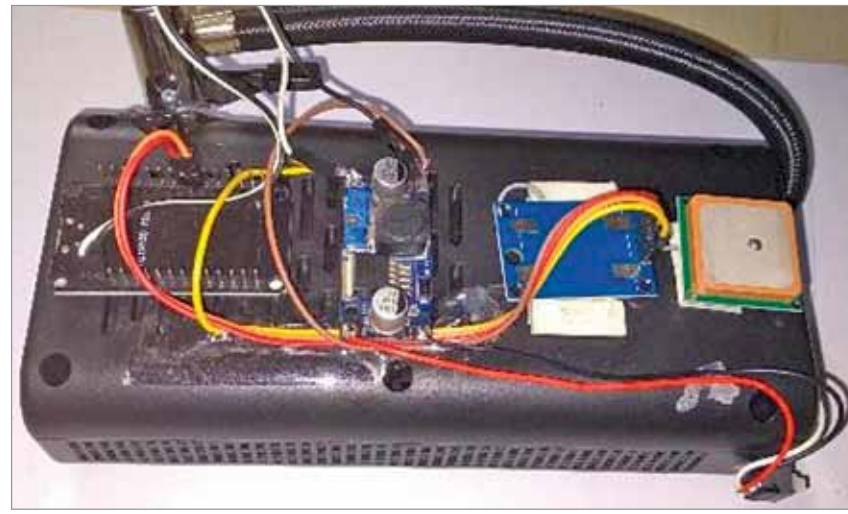


Fig. 1: Authors' prototype

BILL OF MATERIALS

Component	Value/Type	Quantity	Price (₹)
Pressure sensor	MPXH6115A (analogue output) or similar	1	2000
ESP32 board	NodeMCU-32S	1	400
GPS module	u-blox NEO-6M/7M	1	400
Mini air compressor motor	12V DC	1	Miscellaneous 400
Battery	12V DC, rechargeable	1	
DC-DC buck converter	LM2596 module, set to 5V	1	
Switch	SPST, 2-pin	1	
MOSFET trigger switch module	15A, 400W drive module	1	
Total			**3200**

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